

Vulnerability Assessment Report

2nd September 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2026 to August 2026. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The purpose of the database server is to ensure the user data is stored correctly. In a database the data is stored in a tabular format in form of rows and columns so it is easier to filter out specific data when needed using queries. The security of data on the server is most important as if the user data is leaked then many legal or compliance issues will occur and if the attacker disables the server it will hurt the businesses of clients and users which will damage the companies reputation and finances.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor's Hacker	Obtain sensitive information via exfiltration by the hired hacker.	2	2	4
Cybercriminal	Obtain sensitive information by phishing (or other social engineering techniques) the employees or the supplier.	3	3	9
Disgruntled	A disgruntled system admin may	2	3	6

Employee	<i>use his privilege to leak the confidential information online or to competitors to get his revenge from the company.</i>			
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Approach

Since there are many e-commerce companies, the competition is fierce, so a competitor might do some unethical tricks to gain upper hand in the market. Social Engineering will always remain a vulnerability because there will be new ways or there always be a situation where human emotions can be exploited. In companies it is always a certainty that there will be employees who will be unhappy with the company's treatment of them or they might think that the companies might be favoring certain employees like giving bonuses, using them as examples of hardwork etc. which instead of them it should be the unhappy employees who should be treated that way.

Remediation Strategy

Implementation of MFA, policy of strong passwords in systems of all employees, more depth of security and security hardening should be applied to the server and most importantly the database server. Regular audits and backups. A backup server should be implemented in case of the main one going offline. Least Privilege should be strictly revisited to see if any employee is abusing their position. Awareness about new Social Engineering attacks and new malicious techniques.